

## 10.3 Keyphrase Extraction

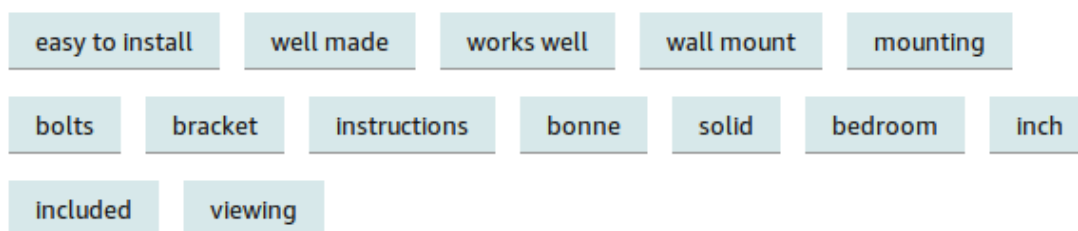
Find the phrases that name what a document is about.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

### 1. The job

A product on Amazon can have a hundred reviews. Nobody reads all of them. Amazon's "Read reviews that mention" filter is **keyphrase extraction** (KPE): find short phrases that many reviewers actually used, then let the shopper click one.

#### Read reviews that mention



KPE is the lightest IE task from note **10.2**. The output is a ranked list of phrases, not a knowledge graph.

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### 2. Supervised or unsupervised

There are two research families.

**Supervised.** You need documents paired with gold keyphrases. Then you train a classifier or a sequence model on engineered features or a network. Labeling is slow and expensive. A model trained on scientific abstracts will not automatically work on tweets.

**Unsupervised.** No phrase-level labels. The method is mostly domain-agnostic. This is what people ship. Recent comparisons have often found that deep supervised KPE does **not** beat a good unsupervised graph method once you count labeling cost.

In production, start unsupervised. Add a short heuristic list for your domain if you have one.

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### 3. Graph-based KPE

The popular unsupervised algorithms share one picture.

1. Turn candidate words and phrases into **nodes**.
2. Put a **weighted edge** between nodes that co-occur (same sentence or a sliding window).
3. Score each node by how well connected it is to the rest of the graph (a PageRank-style walk).
4. Return the top (N) nodes as keyphrases.

A good keyphrase is **frequent enough** and **tied to different parts of the text**. A word that appears ten times in one paragraph and nowhere else is a local echo, not a theme.

Algorithms differ in two places: how they pick candidates, and how they score the graph. Two you will see in code:

| Method   | Idea                                                            |
|----------|-----------------------------------------------------------------|
| TextRank | Rank words like web pages; nearby words vote for each other     |
| SGRank   | Mix statistical cues (position, frequency) with the graph score |

## 4. A small implementation

`textacy` sits on `spacy`. Load English, read a document, then ask for keyterms.

```
import spacy
import textacy

nlp = spacy.load("en_core_web_sm")
doc = textacy.make_spacy_doc(open("nlp_history.txt").read(), lang=nlp)

textacy.extract.keyterms.textrank(doc, topn=10)
textacy.extract.keyterms.sgrank(doc, topn=10)
```

On a short history of NLP, TextRank tends to return long noun phrases (*statistical machine translation system*). SGRank can mix those with shorter, noisier terms (*early, world*). Always **look at the list**. Do not trust the default `topn`.

Overlapping phrases are common: *natural language processing system* and *natural language system*.

`textacy.extract.utils.aggregate_term_variants` groups variants so you can keep one phrase per group.

## 5. Practical advice

Graph methods misbehave in predictable ways.

**Document length.** Building every n-gram and a graph for a 50-page PDF is slow and noisy. A common fix is to use the first (M%) and the last (N%) of the text. Introductions and conclusions usually state the point.

**Overlap.** Each phrase is ranked on its own, so *buy back stock* and *buy back* both survive. Drop near-duplicates with cosine similarity, or use the variant-aggregation helper above.

**Ugly patterns.** A keyphrase that starts with a preposition is rarely what you wanted. Filter those patterns after scoring. That is cheaper than retraining.

**Bad text extraction.** KPE cares about sentence boundaries. A PDF or a scan that glues two columns together will invent phrases that never appeared. Clean the text first; post-process the phrase list second.

A custom stack that works in projects: one graph algorithm, plus a short domain stop-list, plus a post-filter for overlap and junk patterns.

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## 6. Practice

1. Why might unsupervised KPE beat a supervised deep model on a new product category?
2. You extract both *battery life* and *long battery life*. What do you do before you show the list to a user?
3. A 40-page report yields 200 candidates and a slow graph. What cheaper input would you try first?